

ASSIGNMENT - 3

PART – A : SNOWFLAKE SETUP

- 1) Create a database named RETAIL_DB.

```
CREATE DATABASE RETAIL_DB;
```

Results (just now)	
Table	Chart Pivot
1 row	83ms
status	
1	Database RETAIL_DB successfully created. 100.0%

- 2) CREATE SCHEMA RETAIL_DB.ORDERS_DEMO;

```
CREATE SCHEMA RETAIL_DB.ORDERS_DEMO;
```

Results (just now)	
Table	Chart Pivot
1 row	37ms
status	
1	Schema ORDERS_DEMO successfully created.

- 3) Create a warehouse named RETAIL_LOAD_WH.

```
CREATE WAREHOUSE RETAIL_LOAD_WH  
WITH  
WAREHOUSE_SIZE = 'XSMALL'  
AUTO_SUSPEND = 60  
AUTO_RESUME = TRUE;
```

Results (just now)	
Table	Chart Pivot
1 row	82ms
status	
1	Warehouse RETAIL_LOAD_WH successfully created.

- 4) Create a role named RETAIL_DATA_ENGINEER_ROLE.

```
CREATE ROLE RETAIL_DATA_ENGINEER_ROLE;
```

Results (just now)	
Table	Chart Pivot
1 row	34ms
status	
1	Role RETAIL_DATA_ENGINEER_ROLE successfully created.

- 5) Create a user named RETAIL_ENGINEER_USER.

```
CREATE USER RETAIL_ENGINEER_USER
PASSWORD = 'Retail@123'
DEFAULT_ROLE = RETAIL_DATA_ENGINEER_ROLE
DEFAULT_WAREHOUSE = RETAIL_LOAD_WH
MUST_CHANGE_PASSWORD = FALSE;
```

Results (just now)	
Table	Chart Pivot
1 row	246ms
id	status
1	User RETAIL_ENGINEER_USER successfully created.

6) Grant required permissions to the role and user.

I) Grant warehouse usage:

```
GRANT USAGE
ON WAREHOUSE RETAIL_LOAD_WH
TO ROLE RETAIL_DATA_ENGINEER_ROLE;
```

ii) Grant database access

```
GRANT USAGE
ON DATABASE RETAIL_DB
TO ROLE RETAIL_DATA_ENGINEER_ROLE;
```

iii) Grant schema access

```
GRANT USAGE
ON SCHEMA RETAIL_DB.ORDERS_DEMO
TO ROLE RETAIL_DATA_ENGINEER_ROLE;
```

iv) Grant create table permission

```
GRANT CREATE TABLE
ON SCHEMA RETAIL_DB.ORDERS_DEMO
TO ROLE RETAIL_DATA_ENGINEER_ROLE;
```

v) Grant role to user

```
GRANT ROLE RETAIL_DATA_ENGINEER_ROLE
TO USER RETAIL_ENGINEER_USER;
```

PART – B : AZURE STORAGE SETUP

1) CREATE STORAGE ACCOUNT

Home

ramjanblobstorage1_1779357762481 | Overview ...

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : ramjanblobstorage1_1779357762... Start time : 5/21/2026, 3:32:52 PM

Subscription : Azure subscription 1 Correlation ID : fe2f0d10-96d0-4173-8bd6-7836d...

Resource group : Resourcegroup1

> Deployment details

✓ Next steps

[Go to resource](#)

2) Create a container named retail-data.

Home > ramjanblobstorage1_1779357762481 | Overview > ramjanblobstorage1

ramjanblobstorage1 | Containers ☆ ...

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	5/21/2026, 3:33:22 PM	Private	Available	-
☐	retail-data	5/21/2026, 3:36:20 PM	Private	Available	-

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

3) Inside the container, create this folder path: raw/orders/ and Upload the initial CSV file: retail_orders_initial.csv

Home > Storage center | Blob Storage > ramjanblobstorage1 | Containers

retail-data ...

Container

Search

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

retail-data > raw > orders

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	retail_orders_initial.csv...	5/21/2026, 3:48:11 PM	Hot (inferred)	Block blob	301 B	Available

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Part - C: Snowflake External Stage

1) Create a file format for CSV.

```
CREATE OR REPLACE FILE FORMAT RETAIL_CSV_FORMAT
TYPE = CSV
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = '';
```

Results (just now)	
Table Chart Pivot	
1 row 50ms	
id	status
1	File format RETAIL_CSV_FORMAT successfully created.

2) Create an Azure storage integration or external stage.

```
CREATE OR REPLACE STORAGE INTEGRATION
RETAIL_AZURE_INTEGRATION
TYPE = EXTERNAL_STAGE
STORAGE_PROVIDER = AZURE
ENABLED = TRUE
AZURE_TENANT_ID = '24b23ea2-47d3-4211-85f7-b54c8fa30f11'
STORAGE_ALLOWED_LOCATIONS = (
'azure://ramjanblobstorage1.blob.core.windows.net/retail-data'
);
```

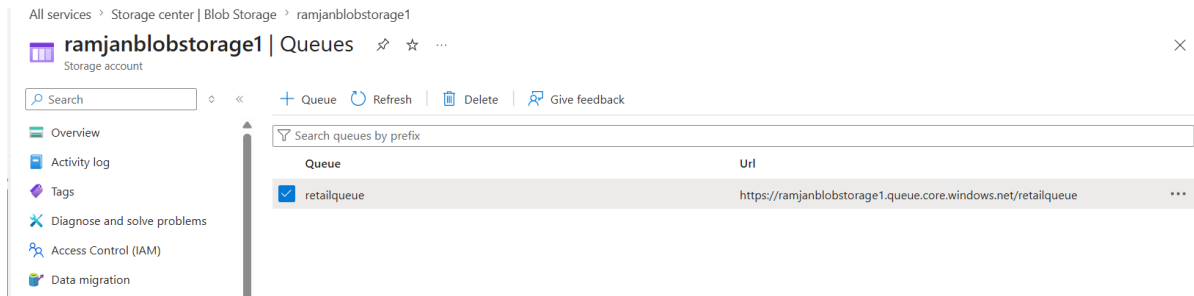
Results (just now)	
Table Chart Pivot	
1 row 92ms	
id	status
1	Integration RETAIL_AZURE_INTEGRATION successfully created.

Adding role assignment :

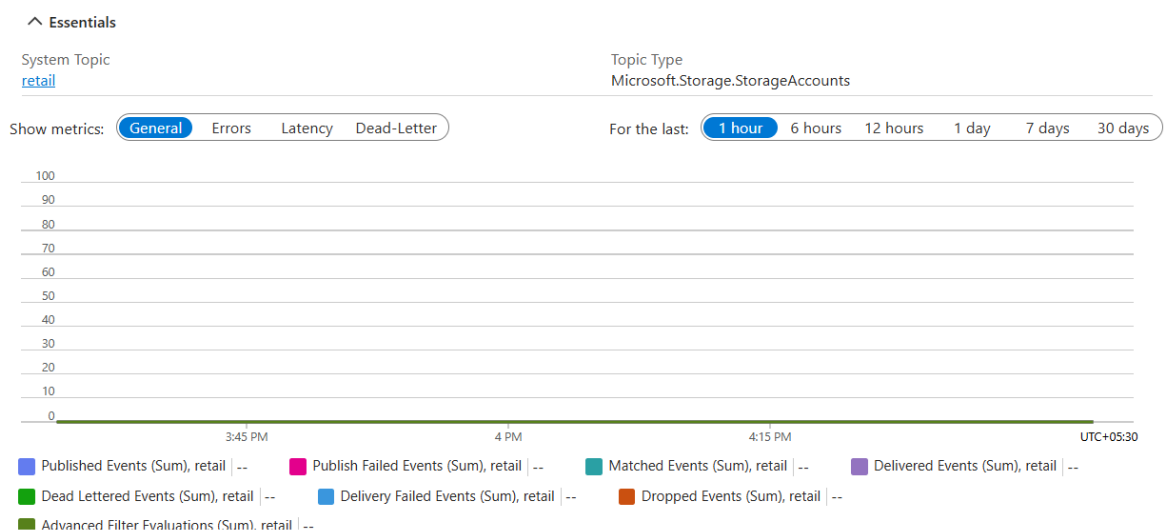
Storage Blob Data Contributor (1)			
☐	 6qaupysnowflakepa... 8fedf729-0a12-4d85...	Service principal	Storage Blob Data Contribut...  This resource
			Add

3) CREATING NOTIFICATION INTEGRATION

I) CREATING QUEUE



II) CREATING EVENT GRID SUBSCRIPTION



III) CREATING NOTIFICATION INTEGRATION

CREATE OR REPLACE NOTIFICATION INTEGRATION

RETAIL_NOTIFICATION_INT

TYPE = QUEUE

ENABLED = TRUE

NOTIFICATION_PROVIDER = AZURE_STORAGE_QUEUE

AZURE_STORAGE_QUEUE_PRIMARY_URI =
'https://ramjanblobstorage1.queue.core.windows.net/retailqueue'

AZURE_TENANT_ID = '24b23ea2-47d3-4211-85f7-b54c8fa30f11';

	status
1	Integration RETAIL_NOTIFICATION_INT successfully created.

Adding role assignment :

Storage Queue Data Contributor (1)			
☐	 o37a13snowflakepac... Odd98736-b43e-429...	Service principal	Storage Queue Data Contrib... This resource
Showing 1 - 3 of 3 results.			

4) CREATING EXTERNAL STAGE

CREATE OR REPLACE STAGE RETAIL_ORDERS_STAGE

URL='azure://ramjanblobstorage1.blob.core.windows.net/retail-data/raw/orders/'

STORAGE_INTEGRATION = RETAIL_AZURE_INTEGRATION

FILE_FORMAT = RETAIL_CSV_FORMAT;

Results (1 minute ago)

TableChartPivot

🔍🗒️1 row ⓘ76ms

↶↷⌵⌴⌚

🗄️	A status
1	Stage area RETAIL_ORDERS_STAGE successfully created.

5) List the files inside the stage

LIST @RETAIL_ORDERS_STAGE;

Results (just now)				
Table	Chart	Pivot		
	name	size	md5	last_modified
1	azure://ramjanblobstorage1.blob.core.windows.net/retail-data/raw/orders/retail_orders_initial.csv.csv	301	6e70e85fcd602280667df	Thu, 21 May 2026 10:18:1

PART – D : Initial Data Load

1) Create a raw landing table named ORDERS_RAW.

CREATE OR REPLACE TABLE ORDERS_RAW

```
(
ORDER_ID NUMBER,
CUSTOMER STRING,
PRODUCT STRING,
CITY STRING,
PAYMENT_MODE STRING
);
```

Results (just now)	
Table	Chart Pivot
1 row	198ms
1	Table ORDERS_RAW successfully created.

2) Create a final table named ORDERS_FINAL.

```
CREATE OR REPLACE TABLE ORDERS_FINAL
(
ORDER_ID NUMBER,
CUSTOMER STRING,
PRODUCT STRING,
CITY STRING,
PAYMENT_MODE STRING
);
```

Results (just now)	
Table	Chart Pivot
1 row	169ms
1	Table ORDERS_FINAL successfully created.

3) Load the initial file into ORDERS_RAW.

```
COPY INTO ORDERS_RAW
FROM @RETAIL_ORDERS_STAGE/retail_orders_initial.csv
FILE_FORMAT = (FORMAT_NAME = RETAIL_CSV_FORMAT);
```

Results (just now)								
	file	status	# rows_parsed	# rows_loaded	# error_limit	# errors_seen	first_error	# first_error_J
1	azure://ramjanblobstor	LOADED	8	8	1	0	null	

4) Insert data into ORDERS_FINAL.

```
INSERT INTO ORDERS_FINAL
SELECT * FROM ORDERS_RAW;
```

Results (just now)	
	# number of rows inserted
1	8

Part E: creating pipeline

```
CREATE OR REPLACE PIPE RETAIL_ORDERS_PIPE
AUTO_INGEST = TRUE
INTEGRATION = RETAIL_NOTIFICATION_INT
AS
COPY INTO ORDERS_RAW
FROM @RETAIL_ORDERS_STAGE
FILE_FORMAT = (FORMAT_NAME = RETAIL_CSV_FORMAT);
```

Results (just now)	
	status
1	Pipe RETAIL_ORDERS_PIPE successfully created.

2) creating stream

```
CREATE OR REPLACE STREAM ORDERS_RAW_STREAM
ON TABLE ORDERS_RAW;
```


Results (just now)

TableChartPivot

1 row

285ms

status

1

Stream ORDERS_RAW_STREAM successfully created.

3) Create a task to merge new rows from the stream into ORDERS_FINAL.

```
CREATE OR REPLACE TASK ORDERS_MERGE_TASK
```

```
WAREHOUSE = RETAIL_LOAD_WH
```

```
SCHEDULE = '1 MINUTE'
```

```
AS
```

```
MERGE INTO ORDERS_FINAL T
```

```
USING
```

```
(
```

```
SELECT *
```

```
FROM ORDERS_RAW_STREAM
```

```
) S
```

```
ON T.ORDER_ID = S.ORDER_ID
```

```
WHEN MATCHED THEN
```

```
UPDATE SET
```

```
T.CUSTOMER = S.CUSTOMER,
```

```
T.PRODUCT = S.PRODUCT,
```

```
T.CITY = S.CITY,
```

```
T.PAYMENT_MODE = S.PAYMENT_MODE
```

```
WHEN NOT MATCHED THEN
```

```
INSERT
```

```
(
```

```
ORDER_ID,
```

```
CUSTOMER,
```

```
PRODUCT,  
CITY,  
PAYMENT_MODE  
)  
VALUES  
(  
S.ORDER_ID,  
S.CUSTOMER,  
S.PRODUCT,  
S.CITY,  
S.PAYMENT_MODE  
);
```

Results (just now)

Table

Chart

Pivot

🔍

🗑️

1 row ⓘ

52ms

🏠

⬇️

🕒

📄	🔗 status
1	Task ORDERS_MERGE_TASK successfully created.

4) UPLOADING THE NEW FILE ORDERS.CSV

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

retail-data > raw > orders

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 3 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	[-]						...
☐	orders.csv	5/21/2026, 6:06:14 PM	Hot (Inferred)	Block blob	272 B	Available	...
☐	retail_orders_incremen...	5/21/2026, 5:56:04 PM	Hot (Inferred)	Block blob	212 B	Available	...
☐	retail_orders_initial.csv...	5/21/2026, 3:48:11 PM	Hot (Inferred)	Block blob	301 B	Available	...

5) Verify that new rows are automatically loaded into Snowflake.

Results (13 minutes ago)

TableChartPivot

🔍🔗

20 rows

77ms

📄⬇️🕒

	# ORDER_ID		A CUSTOMER		A PRODUCT		A CITY		A PAYMENT_MODE	
	120		Kiran Priya +16 more		Headphones Keyboard +13 more		Hyderabad Bangalore +6 more		UPI Credit Card +3 more	
7		2	John		Mouse		Bangalore		Credit Card	
8		3	Asha		Keyboard		Chennai		Debit Card	
9		4	Priya		Monitor		Mumbai		Cash	
10		5	David		Printer		Pune		UPI	
11		6	Kiran		Tablet		Delhi		Net Banking	